

Practice Problem Sheet-1

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Transform Calculus (MA 20202)

1. Find the L.T. of the following functions;

(i) $t e^{-2t} \sin 2t \sin 3t$ Ans: $\frac{1}{2} \left[\frac{(s+2)^2 - 1}{\{(s+2)^2 + 1\}^2} + \frac{25 - (s+2)^2}{((s+2)^2 + 25)^2} \right]$

(ii) $t^2 e^{-t} \cosh 2t$ Ans: $\left[\frac{2(s+1)\{(s+1)^2 + 12\}}{\{(s+1)^2 - 4\}^3} \right]$

(iii) $\frac{e^{at} - \cos bt}{t}$ Ans: $\ln \frac{\sqrt{s^2 + b^2}}{s-a}$

(iv) $\frac{\sin^2 t}{t}$ Ans: $\frac{1}{2} \ln \frac{\sqrt{s^2 + 4}}{s}$

(v) $\frac{\sin 3t \cos 2t}{t}$ Ans: $\frac{1}{2} \left[\pi - \tan^{-1} \frac{s}{5} - \tan^{-1} s \right]$

(vi) $e^{-t} \int_0^t t \cos t \, dt$ Ans: $\frac{s^2 + 2s}{(s+1)(s^2 + 2s + 2)^2}$

(vii) $e^{-t} \int_0^t \frac{\sin t}{t} \, dt$ Ans: $\frac{\cot^{-1}(s+1)}{s+1}$

(viii) $\int_0^t \frac{e^{-t} \sin t}{t} \, dt$ Ans: $\frac{\cot^{-1}(s+1)}{s}$

(ix) $e^{-3t} \int_0^t t \sin^2 t \, dt$ Ans: $\frac{1}{2(s+3)^3} \left[\frac{4 - (s+3)^2}{2(s+3)(s^2 + 6s + 13)^2} \right]$

(x) $e^{4t} \int_0^t \frac{\sin 3t \cos 2t}{t} \, dt$ Ans: $\frac{1}{2(s-4)} \left[\pi - \tan^{-1} \frac{s-4}{5} - \tan^{-1}(s-4) \right]$

The End
